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ARTICLE

Body composition, energy expenditure and physical activity

Comparison of bioimpedance spectroscopy and dual energy X-ray absorptiometry for assessing body composition changes in obese children during weight loss

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Abstract

Background Obesity and age influence the reliability of dual energy X-ray absorptiometry scanning (DEXA) and bioimpedance spectroscopy (BIS). Both are used in clinical settings, but have not been compared for measurements in obese children. We compared DEXA and BIS for evaluating body composition and inherent changes in obese children before and after a 10-month weight loss programme.

Methods DEXA and BIS were used to evaluate 130 patients at baseline and 75 at follow-up. We tested agreement between the two techniques using Bland–Altman plots and proportional bias using Passing–Bablok regressions.

Results The Bland–Altman plots showed wide agreement limits before and after weight loss and when monitoring longitudinal changes. At baseline, the Passing–Bablok regressions revealed a proportional bias for all body compartments. After significant weight loss no proportional bias was found for fat mass and percentage, although BIS systematically underestimated fat mass by 2.9 kg. Longitudinally, no proportional bias was found in the measured changes of absolute fat, fat-free mass and fat-free percentage between both methods, although BIS systematically underestimated fat and fat-free mass by 2.6 and 0.7 kg, respectively.

Conclusion While BIS and DEXA are not interchangeable at baseline, the agreement between the two improved after significant weight loss. Proportional changes in fat mass, fat-free mass and fat-free percentage were similar for both techniques. BIS is a viable alternative to DEXA for future paediatric obesity studies measuring treatment effect at group levels, but is not superior to DEXA and cannot be used for monitoring individual changes due to wide limits of agreement.

Introduction

The obesity pandemic is growing and has become a serious public health challenge [1]. Body Mass Index (BMI), although widely used, has low sensitivity and misses obesity diagnosis in one in four children. Therefore, the interest of clinicians in body composition is increasing [2, 3].

The dual energy X-ray absorptiometry (DEXA) scan is a regularly used method to assess body composition [2], but it has certain limitations. It is expensive, requires the patient to lie still for 20 min, is not portable and exposes the patient to a low dose of radiation. In obese patients, an extra limitation can be present, which is that the body surface of the patient can exceed the scanning area [4, 5]. DEXA has been found to overestimate fat mass and fat percentage in obese children and adolescents as compared to a 4-compartment model [6]. The 4-C model is considered the gold standard, but it is expensive, time-consuming and therefore unfeasible in daily clinical practice [7]. This establishes a need for accurate and reliable techniques for measuring body composition that are inexpensive and suitable for bedside investigation. The Body Composition Monitor® (BCM, Fresenius Medical Care, Germany) that uses bioimpedance

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spectroscopy (BIS), might meet these needs. It is a safe, easy to use, portable device and is relatively inexpensive. Comparison studies of BIS and DEXA in obese adults suggest that high body fat decreases the comparability between both methods [8]. BCM measurements based on BIS are valid over a wide range of body compositions [9]. The BCM integrates a volume model with a body composition model and has been validated against DEXA for the measurement of fat and fat-free mass in healthy adults and adults with renal, liver or oncological diseases [10]. In children, Dasgupta et al. have validated the volume model for determination of fluid overload and normally hydrated weight [11]. However, studies comparing the BIS-derived body composition model with DEXA in a paediatric population, with a longitudinal follow-up, have not been conducted. Given the prediction by Matthie et al. regarding BIS becoming the standard for determining fat percentage and fat-free mass of (morbidly) obese subjects, the validity of BIS needs to be examined in an obese paediatric population [12]. Our objective is to compare DEXA and BIS in an obese population aged 8–18 years before and after a weight loss programme to evaluate whether BIS can accurately track longitudinal changes in body composition.

Materials and methods

Study population and design

A prospective cohort study was performed, including obese children and adolescents aged 8–18 years. Patients were excluded in case of an endocrine or syndromal origin of obesity, genetic syndromes, acute illness, uncontrolled chronic disease or the use of anti-inflammatory drugs.

All patients participated in a 10-month inpatient treatment programme at a paediatric rehabilitation centre 'Het Zeepreventorium' (De Haan, Belgium). The programme aims at weight reduction by restricting dietary intake (1400–1600 kcal/day), increasing physical activity with a minimum of 10 h per week under medical supervision and individual and group psychological support. All patients were asked to complete a baseline visit at the moment of admission and a second visit after 10 months of treatment. All measurements were performed at both visits.

Anthropometry

Height was measured to the nearest millimetre using a calibrated yard stick and weight was recorded to the nearest 0.1 kg using calibrated Seca 899 weighing scales (Seca ltd., Birmingham, England). BMI was calculated as weight (in kilograms) over height (in metres) squared and further analysed as standard deviation scores (SDS) using Flemish growth charts as a reference population [13]. Overweight and obese were defined as per the International Obesity Task Force criteria [14].

Dual energy X-ray absorptiometry

A DEXA scan (Lunar Prodigy Advance, utilising a narrow-angle fan beam technology, encore paediatric software: version ENC V13.6, GE Healthcare, Madison, WI, USA) was performed by a trained technician. As previously shown, this device has high reproducibility [15, 16]. Patients were scanned lying supine and wearing only their underwear. Patients were non-fasting. Data on fat mass, fat percentage, fat-free mass and fat-free percentage were obtained. Fat-free mass was defined as lean tissue mass plus the bone mineral content.

The scanned images were scored based on quality of the study and more specifically, whether the scanned image covered the complete body surface. A scan was considered incomplete when parts of the upper arm fell out of the image, even though these missing values were extrapolated from measurements performed on the other side [17]. Complete scans were scored as zero and incomplete scans as one.

Bioimpedance spectroscopy

A BCM (Fresenius Medical Care, St. Wendel, Germany) measurement was performed the morning after an overnight fast, with patients lying supine. Electrodes were attached following the wrist–ankle approach in a tetrapolar arrangement with two electrodes placed on the hands and two on the feet [18]. Age, gender, height, weight and blood pressure were registered by the device before starting the measurement. If the quality calculated by BCM was <75%, the measurement was repeated and only good quality measurements were used. Data on fat mass, fat percentage, fat-free mass and fat-free percentage were collected.

Statistics

Normality was tested using a Kolmogorov–Smirnov test. Normally distributed data are presented as mean and standard deviation (SD). Skewed data are reported as median and range (minimum–maximum).

Groups were compared using the paired *t*-test for normally distributed data and the Wilcoxon signed-rank test for skewed data. Correlations were calculated using Pearson's or Spearman's correlation analysis. A p value ≤ 0.05 was accepted as statistically significant. The agreement between both measurement methods was evaluated using Bland–Altman plots [19]. Passing–Bablok regressions were used to investigate proportional bias between both techniques [20]. The between-method difference for each compartment at baseline was calculated and

the absolute value of this difference was taken to identify predictors of non-agreement between both methods using linear regression models.

Statistical analysis was performed using Statistical Package for Social Sciences (SPSS) version 25 (SPSS, NY, USA) [21] and XLSTAT (XLSTAT, Addinsoft, NY, USA) [22].

Results

Baseline assessment

One hundred and thirty patients with a mean BMI SDS of 2.7 (range 1.6–3.7) were included in the study. The mean age was 14.4 ± 2.2 years, and 40% of patients were male. Patient characteristics at baseline and follow-up are shown in Table 1.

A strong correlation was found between fat and fat-free mass measured by both methods ($r = 0.963$, standard error [SE] 0.022 and $r = 0.755$, SE 0.058, respectively, both $p < 0.01$). A moderate correlation was present for fat percentage and fat-free percentage ($r = 0.668$, SE 0.066 and $r = 0.662$, SE 0.092, respectively, both $p < 0.01$).

Bland–Altman plots showed wide limits of agreement for all four parameters of body composition when comparing DEXA with BIS, with a high average bias of 2.584 (95% confidence interval [CI] 1.822–3.346) for fat mass, 4.566 (95% CI 3.810–5.322) for fat percentage, 8.967 (95% CI 7.590–10.344) for fat-free mass and 9.156 (95% CI 8.173–10.139) for fat-free percentage (Fig. 1). Passing–Bablok regressions found a proportional and systematic bias between both methods for all four parameters of body composition (Fig. 2, Table 2).

Age and BMI SDS were associated with change in fat percentage (Δfat%), change in fat-free mass (Δfat-free mass) and change in fat-free percentage (Δfat-free%) (all $p < 0.05$). Age also correlated with change in fat mass (Δfat mass), while BMI SDS did not ($p = 0.9$). Female patients had a greater difference in fat and fat percentage between DEXA and BIS as compared to male patients (4.86 ± 2.82 kg and 6.11 ± 3.58% in females as compared to 3.49 ± 2.31 kg and 4 ± 2.89% in males, both $p < 0.01$). Δfat-free mass and Δfat-free% did not differ between genders ($p = 0.2$).

A final linear regression model for the between-method difference (delta values) of all parameters is shown in Table 3. The combination of gender, age, BMI SDS and completeness of the scanned image results in an explained proportion of the variance of 14.6% for fat mass, 40.5% for fat percentage, 62.2% for fat-free mass and 61.0% for fat-free percentage. The interaction between gender and BMI SDS was a significant contributor to the model of Δfat-free mass. Therefore, the linear regression model for the between-method difference in fat-free mass has also been made separately for boys and girls, accounting for the proportion of the variance of 72.9% in boys and 57.9% in girls (Table 4).

Follow-up assessment

Seventy-five patients participated in the follow-up visit, implying a drop-out rate of 42%. This was attributable to premature finishing of the programme by 54 participants and insufficient cooperation during the BIS measurement by one participant. At the time of the follow-up, 12 patients had reached a normal weight, 36 remained overweight and 27 were still classified as obese. On average, the BMI of patients had

Table 1 Patient characteristics before and after weight loss treatment.

	Baseline	After 10 months of treatment	*p* value
N	130	75	
Gender (♂/♀)	52/78	25/50	0.342[1]
Age (years)	14.4 ± 2.2	15.3 ± 2.3	<0.001[2]
BMI SDS (*z*-score)	2.7 (1.6–3.7)	1.8 (0.5–2.7)	<0.001[3]
DEXA fat mass (kg)	46.3 (22.3–88.1)[a]	27.2 (8.9–49.3)[b]	<0.001[3]
DEXA fat percentage (%)	48.9 ± 4.6[b]	35.8 ± 8.7[b]	<0.001[2]
DEXA fat-free mass (kg)	51.9 ± 12.0[a]	49.2 ± 10.9[b]	0.317[2]
DEXA fat-free percentage (%)	51.6 ± 4.6[b]	64.2 ± 8.7[b]	<0.001[2]
BIS fat mass (kg)	43.1 (14.4–93.8)	24.5 (8.1–53.8)	<0.001[3]
BIS fat percentage (%)	43.9 ± 5.8	32.7 ± 8.2	<0.001[2]
BIS fat-free mass (kg)	41.8 (24.9–67.9)	41.5 (27.4–71.4)	0.019[3]
BIS fat-free percentage (%)	42.4 ± 7.5	56.1 ± 10.8	<0.001[2]

Results are presented as mean ± SD or median (minimum–maximum).

[1]Chi-square test, [2]Paired *t*-test, [3]Wilcoxon sign rank test.

[a]$p < 0.05$ compared to the BCM measurement at the same time point, wilcoxon sign rank test.

[b]$p < 0.05$ compared with the BCM measurement at the same time point, paired *t*-test.

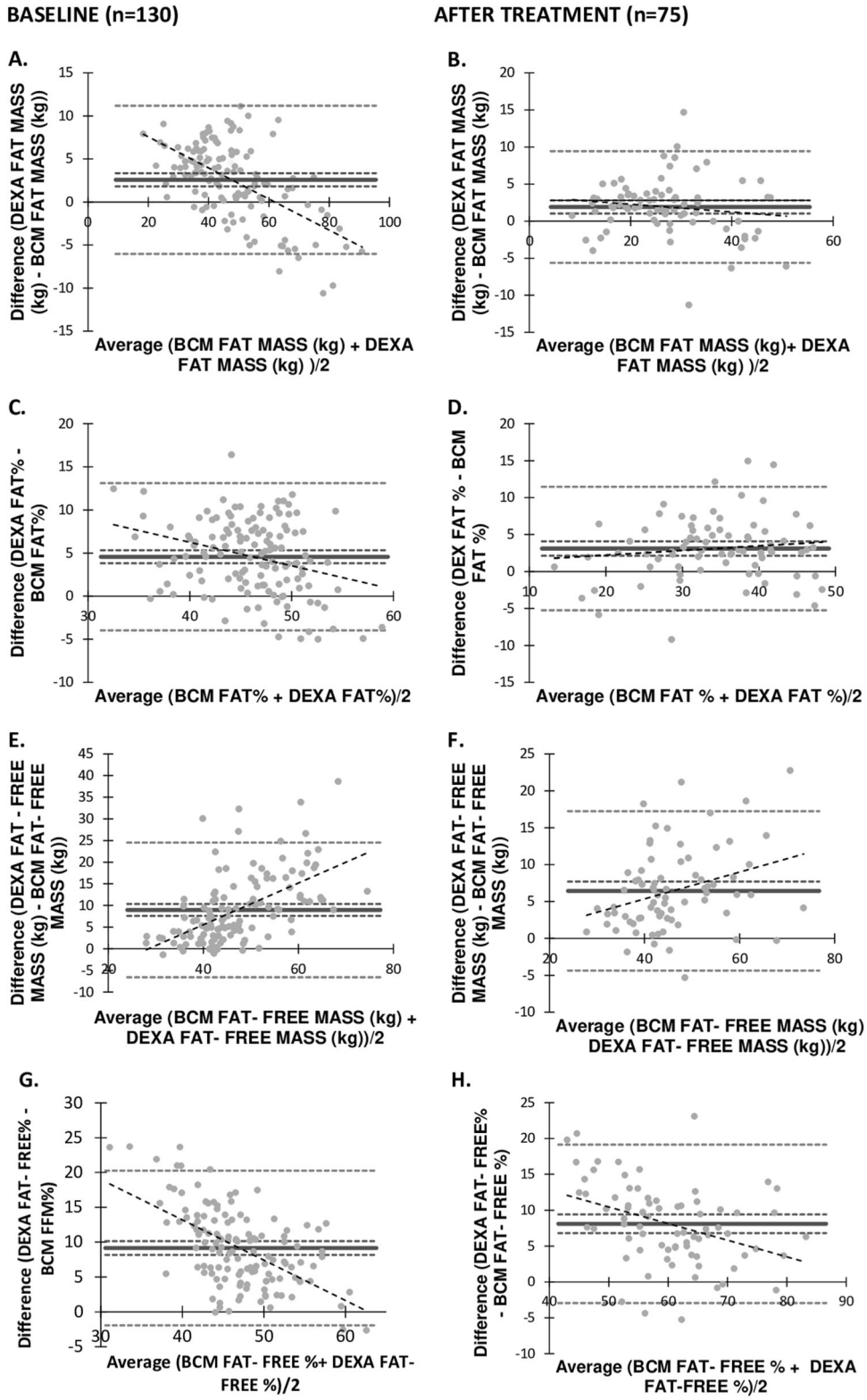

Fig. 1 Bland–Altman plots comparing the BCM- and DEXA-derived measurements of the four body compartments before and after treatment. Fat mass before (**a**) and after treatment (**b**), fat percentage before (**c**) and after treatment (**d**), fat-free mass before (**e**) and after treatment (**f**) and fat-free percentage before (**g**) and after treatment (**h**). The blue solid line indicates the mean difference and the blue dashed lines the 95% confidence interval (CI) around the mean. The red dashed lines indicate the mean ± 1.96 SD. The black dashed line represents the linear regression line of difference. BCM body composition monitor, DEXA dual energy x-ray absorptiometry.

significantly reduced by 1.0 ± 0.4 SDS. Table 1 shows the evolution of the parameters measured by BIS and DEXA.

Baseline characteristics of age ($p = 0.5$), gender ($p = 0.07$) or BMI SDS ($p = 0.5$) of those who came for the follow-up did not differ from those that dropped out. However, patients who dropped out had a higher weight as compared to those who did not (mean 108.57 ± 25.86 and 99.39 ± 22.95 kg, respectively; $p = 0.04$). Those who

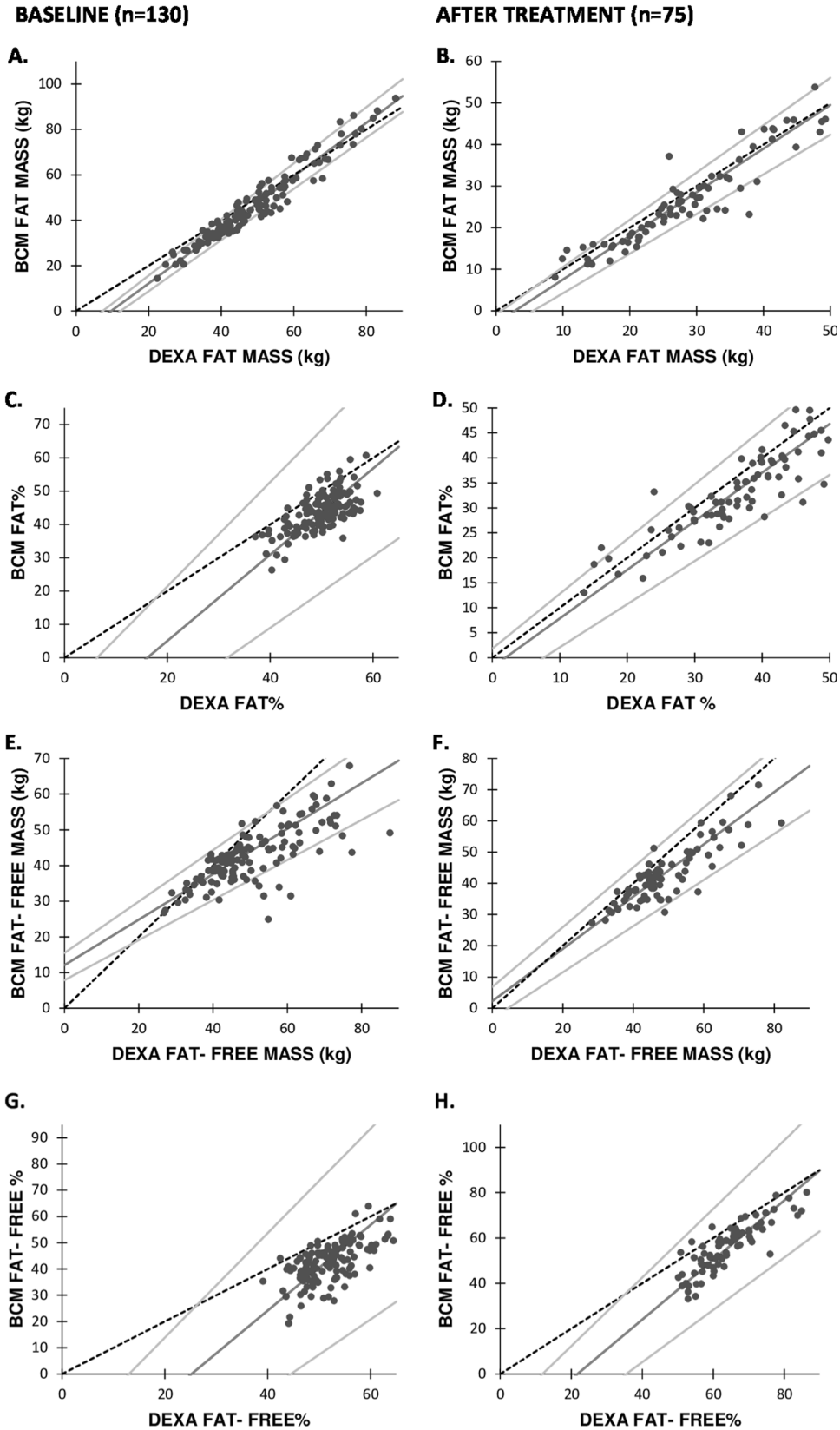

Fig. 2 Passing–Bablok regressions comparing the BCM- and DEXA-derived measurements of the four body compartments before and after treatment. fat mass before (**a**) and after treatment (**b**), fat percentage before (**c**) and after treatment (**d**), fat-free mass before (**e**) and after treatment (**f**) and fat-free percentage before (**g**) and after treatment (**h**). The black dashed line indicates the regression in case of perfect agreement. The red solid line represents the actual regression obtained by the comparison of DEXA with BIS. The grey solid lines represent the 95% CI around the obtained regression. BCM body composition monitor, DEXA dual energy X-ray absorptiometry.

Table 2 Passing–Bablok regression coefficients comparing body composition measured by bioimpedance spectroscopy (BIS) and dual energy x-ray absorptiometry (DEXA).

Moment	Parameter		Value	Lower bound 95% CI	Upper bound 95% CI
Baseline	Fat mass (kg)	Intercept	−11.431	−13.967	−9.143
		Slope	1.180	1.129	1.236
	Fat mass % (%)	Intercept	−20.128	−32.075	−9.051
		Slope	1.317	1.092	1.567
	Fat-free mass (kg)	Intercept	10.061	6.648	13.929
		Slope	0.649	0.569	0.724
	Fat-free % (%)	Intercept	−51.751	−70.787	−37.153
		Slope	1.828	1.540	2.198
After treatment	Fat mass (kg)	Intercept	−2.946	−5.147	−0.977
		Slope	**1.049**	0.957	1.137
	Fat % (%)	Intercept	**−1.870**	−6.542	1.825
		Slope	**0.973**	0.863	1.096
	Fat-free mass (kg)	Intercept	**2.251**	−3.299	6.793
		Slope	0.837	0.740	0.958
	Fat-free % (%)	Intercept	−28.579	−40.913	−17.989
		Slope	1.314	1.153	1.514
Change over time	Fat mass (kg)	Intercept	−2.603	−5.211	−0.722
		Slope	**1.079**	0.930	1.222
	Fat % (%)	Intercept	**0.129**	−1.611	1.628
		Slope	0.833	0.710	0.991
	Fat-free mass (kg)	Intercept	−0.691	−0.927	−0.455
		Slope	**1.113**	0.769	1.565
	Fat-free % (%)	Intercept	**0.114**	2.422	−1.563
		Slope	**1.045**	0.902	1.244

Results in bold indicate that the slope or intercept is similar to the line of identity. (e.g., slope equals 1; intercept equals 0).

dropped out also had a higher DEXA-measured fat-free mass than those who did not (mean 51.17 ± 12.31 and 46.78 ± 10.90 kg, respectively; $p = 0.04$).

The fat mass, fat percentage, fat-free mass and fat-free percentage measured by BIS correlated strongly and significantly with the corresponding DEXA scan measurements ($r = 0.926$, SE 0.043, $r = 0.849$, SE 0.057, $r = 0.805$, SE 0.059, $r = 0.830$, SE 0.061, all $p < 0.01$).

The Bland–Altman plots found wide limits of agreement, although the average bias diminished when compared with baseline for fat mass (1.908; 95% CI 1.024–2.792), fat percentage (3.11; 95% CI 2.130–4.091), fat-free mass (6.436; 95% CI 5.169–7.703) and fat-free percentage (8.103; 95% CI 6.806–9.399) (Fig. 1). No proportional differences were found by the Passing–Bablok regression analysis for fat mass and fat percentage, although BIS underestimated the fat mass by 2.9 kg on average as compared to DEXA (Fig. 2, Table 2).

Female patients had a higher between-method difference for fat percentage as compared to male patients (4.18 ± 4.09% and 0.98 ± 3.85, respectively, $p < 0.01$). No gender differences were found for the between-method differences in fat mass ($p = 0.052$), fat-free mass ($p = 0.256$) and fat-free percentage ($p = 0.455$).

The differences between BCM and DEXA for measurement of fat-free mass and fat-free percentage were lower in normal and overweight participants when compared with obese patients ($p < 0.01$), with an average bias of 3.7 ± 3.3 kg (5.7 ± 5.2%) in normal weight and 4.5 ± 3.5 kg (6.5 ± 4.6%) in overweight patients as compared to 10.4 ± 6.5 kg (11.4 ± 5.9%) in obese patients.

Longitudinal changes in body composition

To assess whether both methods equivalently report longitudinal changes in body composition, the differences between baseline and follow-up were calculated for each parameter. The differences detected in each compartment were also compared between both methods.

As observed earlier, highly significant correlations were obtained for each body compartment measured by both BIS and DEXA (all $p < 0.01$). Strong correlations were found between fat mass ($r = 0.875$, SE 0.055), fat percentage ($r = 0.782$, SE 0.073) and fat-free percentage ($r = 0.788$, SE

Table 3 Linear regression models identifying determinants of the between-method difference at baseline.

Outcome	r	p value	Adjusted R^2
Δ fat mass (kg)			0.146
Age	−0.214	0.016	
Gender	0.182	0.041	
BMI SDS	0.203	0.022	
Complete scan	−0.223	0.012	
Δ fat % (%)			0.405
Age	−0.375	<0.01	
Gender	0.318	<0.01	
BMI SDS	−0.058	0.514	
Complete scan	−0.261	<0.01	
Δ fat-free mass (kg)			0.622
Age	0.549	<0.01	
Gender	−0.251	<0.01	
BMI SDS	0.458	<0.01	
Complete scan	0.102	0.256	
Δ fat-free % (%)			0.610
Age	0.601	<0.01	
Gender	−0.203	0.022	
BMI SDS	0.278	<0.01	
Complete scan	0.210	0.018	

Table 4 Linear regression models identifying determinants of the between-method difference in fat-free mass at baseline for boys and girls separately.

Outcome	r	p value	Adjusted R^2
Δ fat-free mass (kg)			
Boys			0.729
Age	0.432	<0.01	
BMI SDS	0.680	<0.01	
Complete	−0.204	0.156	
Girls			0.579
Age	0.575	<0.01	
BMI SDS	0.334	<0.01	
Complete	0.237	0.04	

0.086). A weak to moderate correlation was found for fat-free mass ($r = 0.439$, SE 0.107) as measured by DEXA and BIS.

The Bland–Altman plots found wide limits of agreement for the changes in fat mass, fat percentage, fat-free mass and fat-free percentage. No systematic bias was found in the measurement of fat-free percentage, with an average bias of 0.607 (95% CI 0.383–1.597). A positive bias was found for fat mass (1.112; 95% CI 0.197–2.028), fat percentage (1.886; 95% CI 0.955–2.818) and fat-free mass (1.693; 95% CI 0.651–2.736). Passing–Bablok regressions analysis showed no proportional bias for fat mass, fat-free mass and fat-free percentage, although BIS underestimated fat mass and fat-free mass (Fig. 3, Table 2).

No gender differences were found in the longitudinal changes measured between both methods.

Discussion

Our findings suggest that DEXA and BIS are not interchangeable methods of measuring body composition in an obese paediatric population. After a weight loss programme, all participants had reduced BMI SDS, but most were still overweight. The agreement between body composition measurements improved at follow-up, but the overall results showed an underestimation of fat mass and an overestimation of fat-free mass by BIS as compared to DEXA.

The results of our baseline and follow-up comparisons are in agreement with a previous study that concluded that BIS (Xitron Hydra 4000B) underestimates fat mass in overweight children as compared to DEXA (Lunar-DPX-L) [23]. Similar results were found in obese adults when comparing BIS (Xitron Hydra 4200 and Impedimed SFB7) to Prodigy DEXA [24–26]. With respect to fat-free mass, the results in literature are more ambiguous. Ellegard et al. found no significant differences between fat-free post-partum mass measurements from obese women using BIS (Xitron Hydra 4200) and Prodigy DEXA, although there were wide limits of agreement at the individual level [24]. Whereas in overweight or obese men, BIS (Impedimed SFB7) overestimated fat-free mass when compared with Prodigy DEXA [25].

Previous studies comparing the BCM with Prodigy DEXA have been conducted in adults and revealed similar between-method differences for fat mass and fat percentage, as seen in our study at the follow-up assessment. They reported an average underestimation of fat mass determined by BIS, ranging from 0.5 to 3.1 kg, as compared to DEXA in healthy adults and adults with liver, renal or oncological diseases [10, 27–30], and an underestimation of fat percentage by 1.78% [29]. Females showed a larger difference in fat percentage estimates than males, which was consistent with our findings [29]. The mean BMI in these studies ranged from 25.4 to 28.0 kg/m^2 which is similar to the BMI of our population at follow-up. Our bias in fat-free mass was much larger than those reported in two studies on dialysis patients, but these studies calculated fat-free mass differently [28, 30]. At follow-up, the bias in fat-free mass and fat-free percentage was significantly lower in normal and overweight patients compared to obese patients. Therefore, BMI seems to be an important confounder. Indeed, the influence of BMI on the agreement between DEXA and BIS has previously been reported in other studies [23, 29, 31, 32].

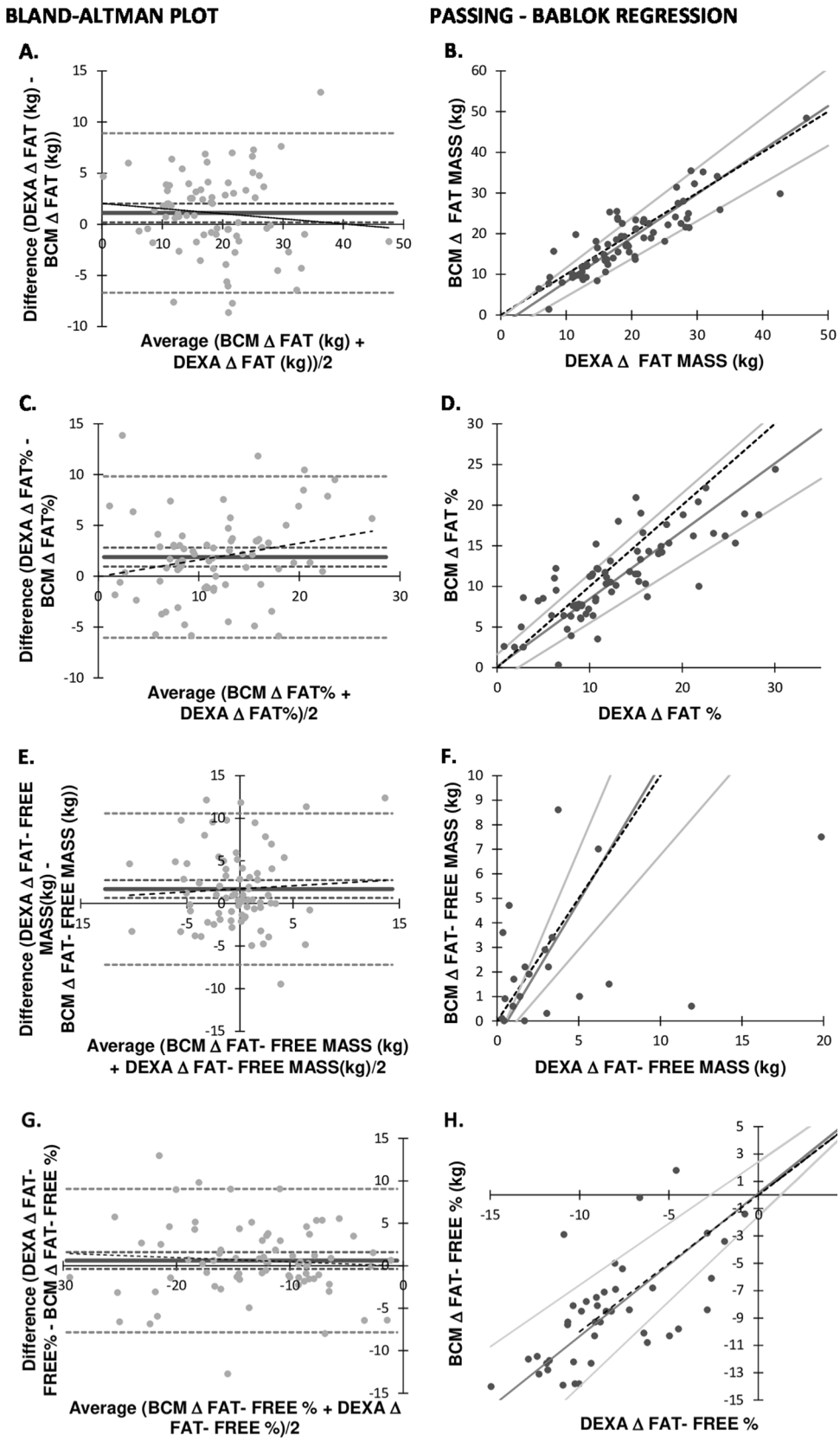

Fig. 3 Bland–Altman plots and Passing–Bablok regressions comparing the longitudinal changes. In the Bland–Altman plot, the blue solid line indicates the mean difference and the blue dashed lines the 95% CI around the mean. The red dashed lines indicate the mean ± 1.96 SD. In the Passing–Bablok regression, the black dashed line indicates the regression in case of perfect agreement. The red solid line represents the actual regression obtained by the comparison of DEXA with BIS. The grey solid lines represent the 95% CI around the obtained regression. The black dashed line represents the linear regression line of difference.

In children of normal weight, the overestimation of fat percentage by Hologic QDR-2000W DEXA, as compared to the 4-C model, varies from 1.13 to 3.9% [33, 34]. In overweight and obese children, the average overestimation of fat percentage measured by a Prodigy DEXA scan, as compared to a 4-C model, was 1.9% (95% limits of agreement 2.1–5.9) [35]. The overestimation of fat percentage by DEXA compared to BIS found in our data, is within the range that DEXA overestimates the 4-Cmeasurements even in normal children, implying that the determined difference at follow-up might be due to an error of DEXA rather than BIS.

Multiple explanations can be found for the lack of agreement at baseline. First, most subjects had an anteroposterior thickness of >20 cm, which amplifies the error in measuring fat and fat-free mass by DEXA due to beam hardening [5, 36, 37]. Second, most individuals had a body surface exceeding the scanning area at baseline. Although the values for missing body parts are estimated by the scanner software based on the fully scanned parts, the reliability of the initial measurements might be low. After weight loss, most of the scanned images covered the entire body surface, which might partly explain the better agreement seen in this case. In contrast with other studies [5], we have incorporated completeness of the scan as a predictor of the between-method differences at baseline. This was found to be significant for estimating the difference in fat mass, fat- and fat-free percentage in both genders and fat-free mass in girls between BIS and DEXA. Third, DEXA and BIS assume a fixed hydration of soft tissues [38, 39]. The hydration keeps declining from 4 years of age until 20 years of age and increases in parallel with increases in adiposity [40, 41]. Moreover, alterations in extracellular water content only result in small effects on the DEXA-derived fat mass [42]. BCM is corrected for BMI, but uses adult hydration references. The use of the adult references might also influence the comparability between both methods, although obesity seems to be the largest contributor to the lack of agreement, as can be inferred by the improved agreement after weight loss and smaller bias in fat-free mass in the normal weight group.

It has to be kept in mind that BCM was originally developed to assess hydration status [43], although recently, it is increasingly being used for nutritional assessment as well [44]. Using BIS, the measured resistance has been implemented in volume and body composition models. The volume model uses programmed software and the mixture theory with anthropometric measurements and certain assumptions to estimate total body water (TBW), intracellular (ICW) and extracellular water (ECW). Lean tissue, adipose tissue and overhydration (OH) are then calculated based on the determined ECW and TBW [9]. We have used fat- and fat-free mass instead, because the adipose tissue mass as reported by BCM also contains adipose water [28]. Individuals with excess adipose tissue have a higher extracellular water content [45] and alterations in body water can influence the DEXA-derived estimates of fat-free mass [46]. This could enhance the bias between both methods if the extra body water is assigned to the fat mass by BCM.

We did not find any proportional biases between BIS and DEXA for the longitudinal changes in fat-free percentage, fat and fat-free mass induced by weight loss. Studies in bariatric surgery patients found that the degree of fat loss was associated with an overestimation of loss of TBW by BIS (Xitron Hydra 4200), caused by rapid and acute fluid shifts elicited by the severe weight loss [47]. Thomson et al. [48] investigated overweight and obese women before and after a 10-week weight loss intervention and reported a good agreement between BIS (Impedimed SFB7) and Lunar Prodigy DEXA in assessing changes in body composition and found no significant differences in the fat percentage, fat and fat-free mass measured. This suggests that BIS might be more accurate in interventions with a slower weight loss as opposed to the rapid weight loss induced by bariatric surgery. In our study, patients were measured after 10 months of weight loss resulting in an average BMI SDS decrease of 1.16 ± 0.33 SDS. Previous studies in the same centre report a mean decrease of BMI SDS ranging from 0.76 to 0.86 SDS after 5 months [49, 50]. This suggests that the highest weight loss takes place in the first few months and that there is still some weight loss at a slower rate during the second part. Following the assumption of Thomson et al. [48], BIS can be used for assessing body composition longitudinally in these sort of interventions where the weight loss occurs more slowly as opposed to surgically-induced weight loss.

A few limitations of our study need to be considered. First, we had a drop-out rate of 42% at the follow-up assessment, which is common in obesity research [51]. Conventional obesity treatment is highly prone to compliance and motivational issues and, therefore, high long-term follow-up rates are difficult to achieve. Since the population lost to follow-up only differed significantly in weight and DEXA-measured fat-free mass, we believe that the remaining follow-up population represented a considerable proportion of the initially included population. A bias between both methods at baseline was also observed only when the group that finished the programme was analyzed. Secondly, as per protocol we performed only one BCM measurement, and a second one only in cases where the quality was <75%. A previous study performed in 20 healthy children using BCM reported high intra- and interrater intraclass correlation coefficients for fat mass and fat-free mass, suggesting limited variability between multiple measurements and technicians [52]. High in vivo reproducibility was also found in a study of Wabel et al. [53]. Third, BCM relies on adult reference values, which is

a limitation when using this equipment in children. If age-specific adipose tissue hydration parameters were introduced, it could extend the applicability to children for determination of excess fluid [43]. Our goal was to evaluate body composition measurements based on ECW and TBW, not OH. Using adult references can induce errors in body composition results since the ECW/ICW ratio might differ in children [40]. Since both measurements were only 10 months apart, we expected a very limited age-related change in hydration status between both assessments, resulting in a similar error in the determination of body composition at both timepoints. Calculating the changes in body composition might overcome this limitation. Apart from age, our follow-up population is very similar to the one used to initially create the model concerning fat percentage, fat-free mass and BMI [43]. The similarity makes this an interesting paediatric population in which to assess the validity of BCM. Fourth, devices and software versions might differ in their body composition estimates. We therefore referred mostly to studies using the same devices for DEXA and BIS measurements, although we used a paediatric DEXA software unlike in adult studies. Besides the technological differences between BIS and DEXA, the software used to analyse the tissue attenuation might be an additional confounder in the agreement between DEXA and BIS. Finally, since there was high individual variability, as shown by the wide limits of agreement on the Bland–Altman plots, these data should not be used for interpretation on an individual level. Nevertheless, when considering the longitudinal results on a group level in an obese paediatric population, BIS might be an interesting technique to use, although both BIS and DEXA have their limitations.

Since there are very limited studies on the usability of BIS to track changes in body composition, these results will need to be confirmed by larger studies. Furthermore, the comparison of BIS with the real gold standard, the 4-C model, in an obese paediatric population would allow for a more definite conclusion. However, this is difficult to achieve in routine clinical practice due to the need for specialised equipment, time and trained technicians to perform these investigations.

The results of this study offer a first step towards implementing BCM in the clinical paediatric obesity research field. This would be of great value since BIS is an inexpensive and simple technique.

Conclusion

The agreement between DEXA and BIS improved after weight loss in obese children and adolescents. However, a trend of underestimation of fat mass and overestimation of fat-free mass by BIS was observed. Both methods report the proportional changes in body composition during weight loss treatment in a similar manner for fat-free percentage and fat- and fat-free mass, although the latter two are systematically underestimated by BIS. These findings suggest that BIS might be an interesting, but not superior, alternative to DEXA for future paediatric obesity studies in which changes in body composition are measured at a group level. However, the results should not be used for monitoring individual changes.

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Author contributions All authors certify that they have participated in the study and take responsibility for the content. The study was designed by MY, AVE, KVH, SV, BDW, LB, MVH and ADG. Measurements were performed by MY, AVE and EV and the database was made by MY, EV and AVE. Data were analysed by EV, AVE under the supervision of SV, who has a Master's degree in Statistics. Data interpretation was done by EV, AVE, SV, KVH, BDW and KVD. All authors were involved in the writing of the manuscript.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval The local ethics committee approved this study (EC n°B670201731779).

Informed consent Written informed consent was obtained from the patients and their parents or legal guardians.

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